Exploratory Analysis on Crime Records in Los Angeles
(January 2020 - April 2025)

Location, Time, and Demographic Observations

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Repository Link

Introduction

The first cases of COVID-19 pandemic were discovered in the United States in December of 2019, but the first widespread effort to contain the disease started a few months later, in March of 2020. What was originally intended to be a short-term lockdown to prevent the spread of the disease, turned into a years-long effort to curb the deadly virus. Many of the experiences we shared during that time seem surreal five years later.

With so many aspects of life changed by COVID-19, our team set out to investigate how crime was also impacted during this period. We selected a dataset of Los Angeles crime data from 2020-2025. The dataset is curated and maintained by the Los Angeles Police Department. It is unclear if the dataset includes Los Angeles City only, or if the data also includes the surrounding county. In total, there are just over 1 million rows and 28 columns in the dataset. Our analysis focused on a few of these variables, namely Date, Area, Crime Code, Age, and Sex.

Our overarching hypothesis was that the COVID-19 Pandemic had a significant impact on crime. After initial analysis our group decided to focus on three questions. Each question compares the COVID time period against the Post-COVID period. For the purposes of this project, the COVID-19 pandemic time period starts in March 2020 and ends in May 2023, which is in line with when the World Health Organization declared an end to the global health emergency.

Unfortunately, with only three months of data prior to March 2020, the dataset was not long enough to make a meaningful analysis of the Pre-COVID period. There are substantially more rows in the COVID period than the Post-COVID period, so we have normalized values so that they can be meaningfully compared. Representing counts of the COVID period versus the Post-COVID period would misrepresent the dataset in this case.

With the above hypothesis in mind, our team set out to explore the following questions:

1. How did crime change by area during the COVID and post-COVID periods?
2. How did crime change between the COVID and post-COVID periods? Did crime change in frequency, type, or time?
3. How did crime against certain victim demographics change between the COVID and post-COVID periods?

Dataset:
https://data.lacity.org/Public-Safety/Crime-Data-from-2020-to-Present/2nrs-mtv8/about_data?no_mobile=true

Exploratory Data Analysis (EDA)

Initial Exploration

Initial exploration of the data showed that the vast majority of the rows in the dataset contained complete data. The exceptions to this finding were in columns that did not apply to the crime report. For example, a third of the reports do not include data about a weapon, because not every crime is commissioned with the aid of a weapon. Overall, we rated the completeness of the dataset very highly. Collectively, there were just over 5,000 duplicates, with no discernible reason for the duplication, so we dropped them.

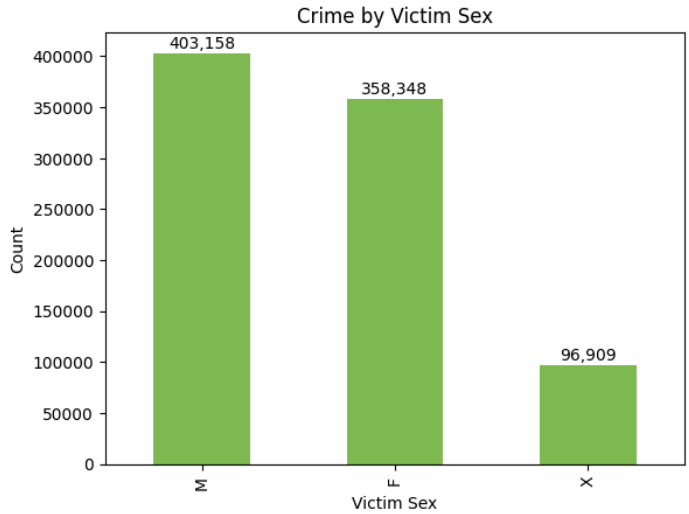


Figure 1: Crime by Victim Sex shows the number of crimes committed against Male, Female and Unknown victims.

Next, we found some peculiarities regarding the victim demographic columns in the data. There are 5 unique values for the column 'vict sex': 'M', 'F', 'X', 'H' and '-', but the dataset references only 'M' 'F' and 'X' where 'X' stands in for an unknown sex. In total, there were only 115 of these undocumented references to 'H' and '-' in the dataset, and we made the assumption that they were erroneous entries. We dropped them when we considered victim sex in our analysis.

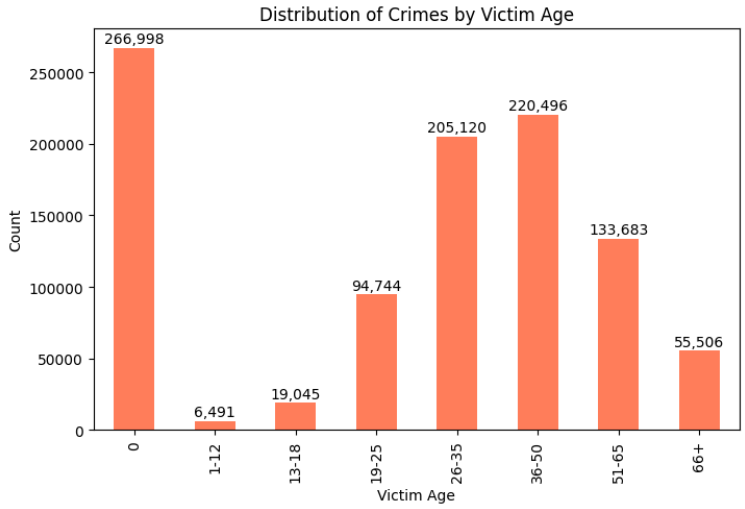


Figure 2: Distribution of Crimes by Victim Age

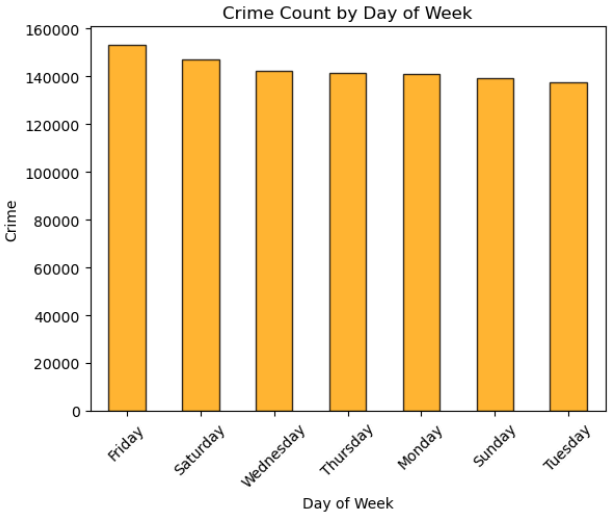
Another noteworthy finding in the analysis was due to the way age was handled in the dataset. As seen in Figure 2, we broke down age into 8 bins ranging from age 0 to 66+. We found that the highest victim age group was from ages 36-50, and the second highest was ages 26-35. We also discovered that more than 260k entries listed the victims age as zero or less. Of those, the overwhelming majority of the entries were zero, while approximately 133 entries were less than zero.

We analyzed the crimes where the victim's age was listed as less than zero and we didn't find anything significant or any correlation between them. We decided that these negative entries were likely erroneous and we dropped them when we considered victim age in our analysis.

For the large number of crimes with a listed victim age of exactly zero, we found that these crimes fit into a few different categories. First, the top 12 entries in this category by count are all variations of theft. The rest of the categories vary, but they fit into a category where a victim's age might not apply, like in the case of discharging a firearm where no one was injured. We ascertain that zero was likely used as a default variable when the crime is "victimless" or otherwise not determined. We kept these entries in the dataset, but factored them in when we analyzed a question regarding age.

Initial Analysis

After we understood the dataset more clearly, we plotted some basic queries from the dataset to answer some initial questions that were related to the larger questions that we were interested in answering.



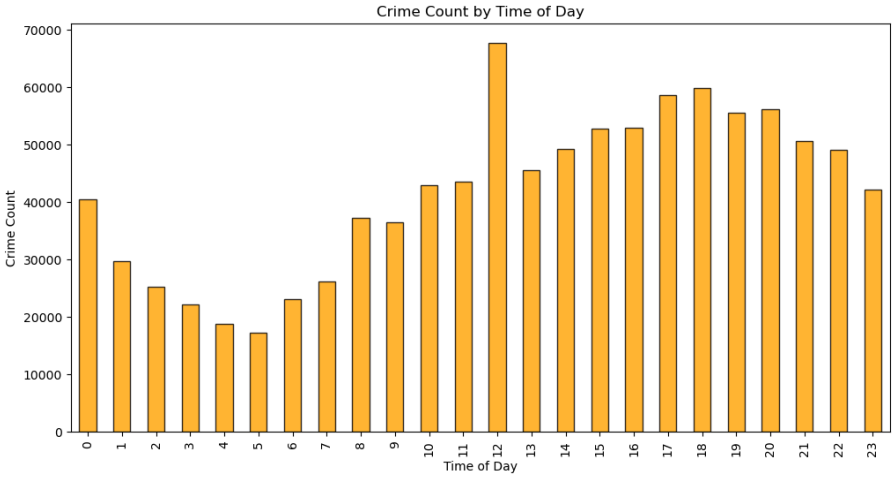


Figure 3 (left):Crime counts by the day of the week, showing some increase over Friday and Saturday compared to the rest of the week.
Figure 4 (right): Crime count by time of day

Before considering questions about how crime changed through the COVID periods, we plotted the following figures to better understand how crime is distributed throughout the week and throughout a day across all periods.

Figure 3 shows crime counts as a function of the day of the week. Observe how slightly more crime is conducted over the weekend, peaking on Friday, but overall crime is relatively well distributed throughout the week.

We find initially that total crime is distributed relatively evenly throughout the week. There may be some crimes that are unevenly distributed throughout the week, but we did not explore this topic. Instead, we reasoned that crime counts as a function of time of day would be more likely to change due to COVID-19.

After the onset of the pandemic, many cities endured loss of employment, hybrid work schedules, and full time remote work altogether. We believed that these facts may lead to a change in the time of day that crime was conducted. Consider for instance, that robberies may now be more difficult to pull off in the middle of the day, because more people are home. To gain a baseline for

future comparison, observe Figure 4.

Across the entire dataset, crime is most likely to be committed at noon, presumably while people are on a lunch break. After noon, total crime increases until it peaks around six in the evening. Our initial assumption is that between 12 and 6 PM is when most people are out of their homes and therefore more vulnerable to crime. While crime peaks around 6 PM, it is notable that there are substantially high crime counts that carry on through the evening until the early morning.

Data Wrangling

This dataset is fairly complete throughout. In terms of data wrangling, we only made a few adjustments to the dataset. First, we made sure that the dates were converted to DateTimeIndex. Between the two date variables, “Date Reported” and “Date Occurred” we kept the occurrence date out of interest for when the crime is projected to occur, rather than it being reported.

We also created tables to easily index between the two periods. If a crime was before May 2023, it was labeled as “During COVID” and “Post-COVID” if it was after that date.

Crime codes were binned into broader categories (e.g., 8 "Personal Theft" codes became one) using Unified Crime Reporting standards for simpler analysis. Major categories included Homicide, Sexual Assault, Robbery, Aggravated Assault, Simple Assault, Burglary, Grand Theft Auto, Burglary Theft from Vehicle, Personal Theft, and Other Theft.

Analysis of Primary Questions

Question 1:

Is the crime number affected from during covid to post-covid?

First we sum the crime count by area and by time period (during covid and post covid) and then divide the count by year as average crime count per year for during covid and post covid. Then we sorted the chart by crime count from during covid, the chart is shown below.

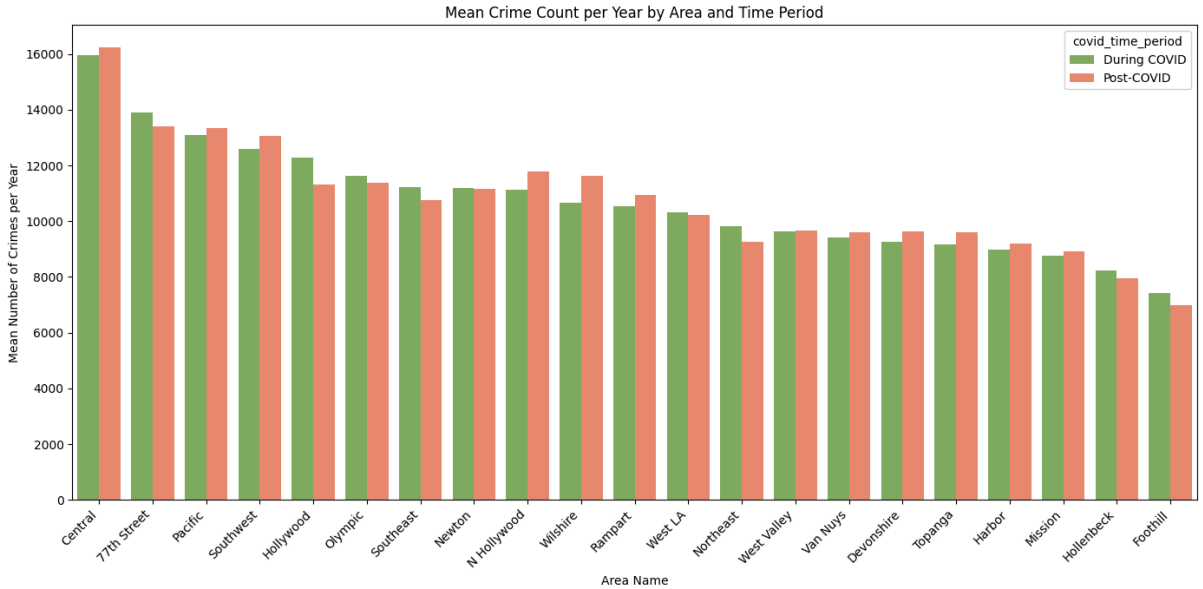


Figure 5: Number of Crimes per year by area for COVID and post-COVID time periods.

Eleven areas experienced a higher average crime count in the post-COVID period, while seven areas saw a decrease in their average crime count. Additionally, three areas maintained a relatively consistent average crime count across both periods. When examining specific areas during and after COVID, Central consistently showed the highest crime count, and Foothill the lowest.

We then drilled down to the percentage changed from during COVID to post-COVID:

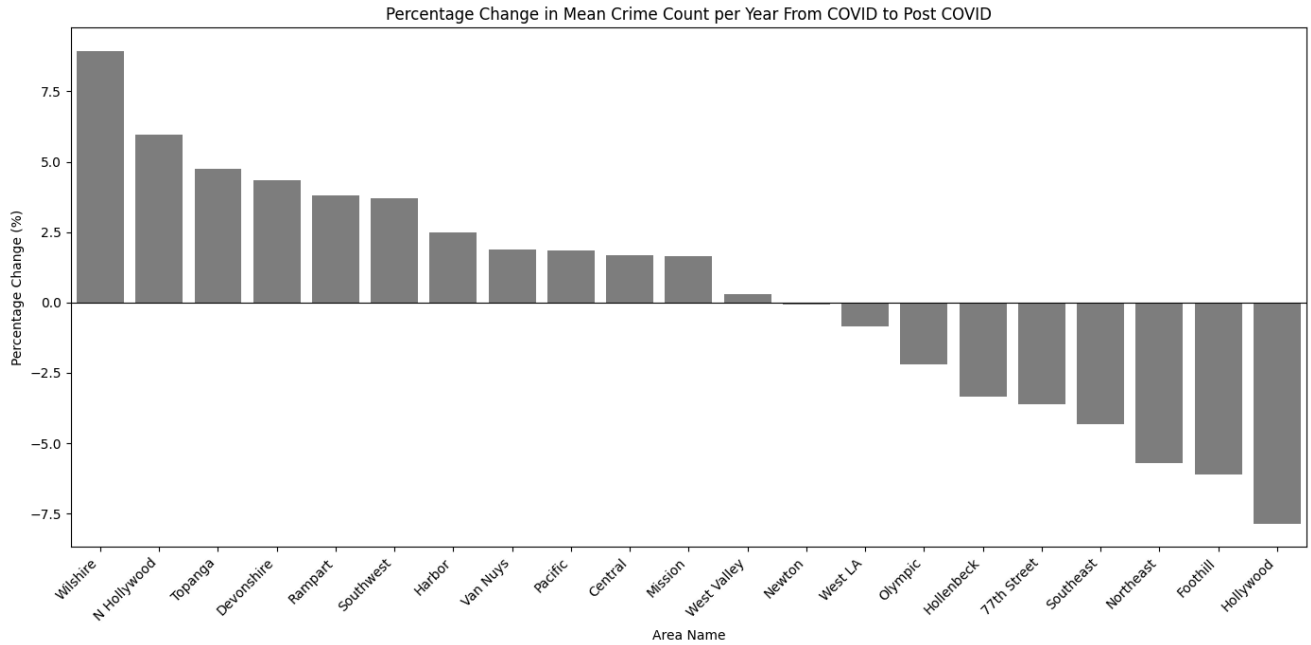


Figure 6: Percentage change in Crime count per year by area for Post-COVID and COVID

Examining the actual percentage change in crime by area reveals that Wilshire experienced the highest percentage increase in crime during the post-COVID period. Conversely, Hollywood saw the most significant decrease in average crime count in the same post-COVID timeframe. Notably, the West Valley, Newton, and West LA areas appear to have been the least affected in terms of crime fluctuations when comparing the pandemic period to the post-COVID period.

Dominant Crime Types Shifts in Each Area based on Count

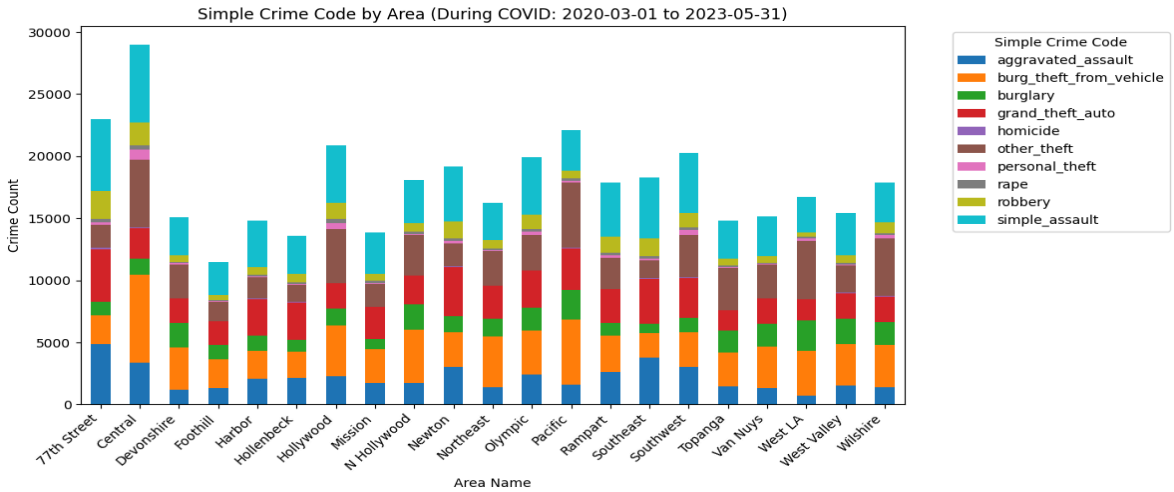


Figure 7: Total simple crime type count during covid, by area.

Figure 7 shows the distribution of crime codes by area. This plot shows us that higher crime areas appear to have proportionally more crimes across all categories. The inverse is true of low crime areas as well. Some of the more serious crimes are not represented in the low crime areas.

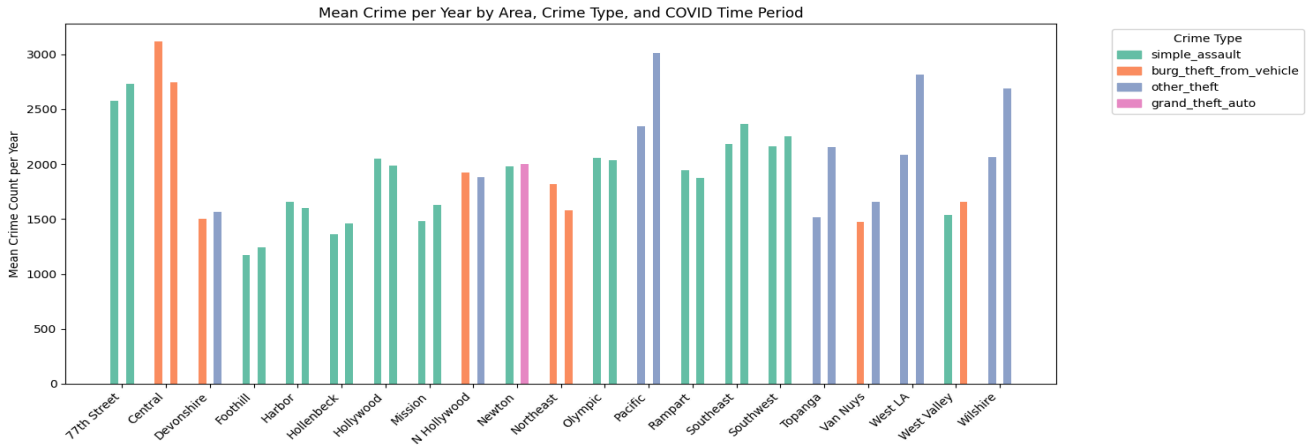


Figure 8: Top 1 mean crime count per year for most frequent crimes, by area.

Analysis of crime trends reveals that simple_assault was a persistent issue spanning both the COVID and post-COVID eras in almost half of the area. A more striking alteration in the crime landscape was the emergence of other_theft as a significantly more common offense in the post-COVID period, signaling a potential shift in criminal focus or opportunity. Talks about the similarity, Central and Northeast shared the same common crime type, burg_theft_from_vehicle, from both time periods.

Pacific, Topanga, WestLA, Wilshire have the most dominant crime type, other_theft, as common crime type in both periods. Notice how the crime type differs from during covid to post covid in the following areas: Devinshire, N Hollywood, Van Nuys, Newton, West Valley. We also found it interesting to note that Grand Theft Auto only appears as a top crime in one reported area, in Newton.

In summary: the dominant crime types only shifted in 5 areas between the two periods. This ran counter to our hypothesis that COVID-19 would change how and where crime was conducted.

Question 2:

How did crime change between the two periods? Did crime change in frequency, type, or time?

Frequency: With COVID changing so many aspects of our lives, this question asks whether or not changes on a societal level impact the types, frequency, or times associated with the conduct of certain crimes. To begin this exploration, observe figure 9, a time series plot of the larger "bins" of crime over the time frame analyzed.

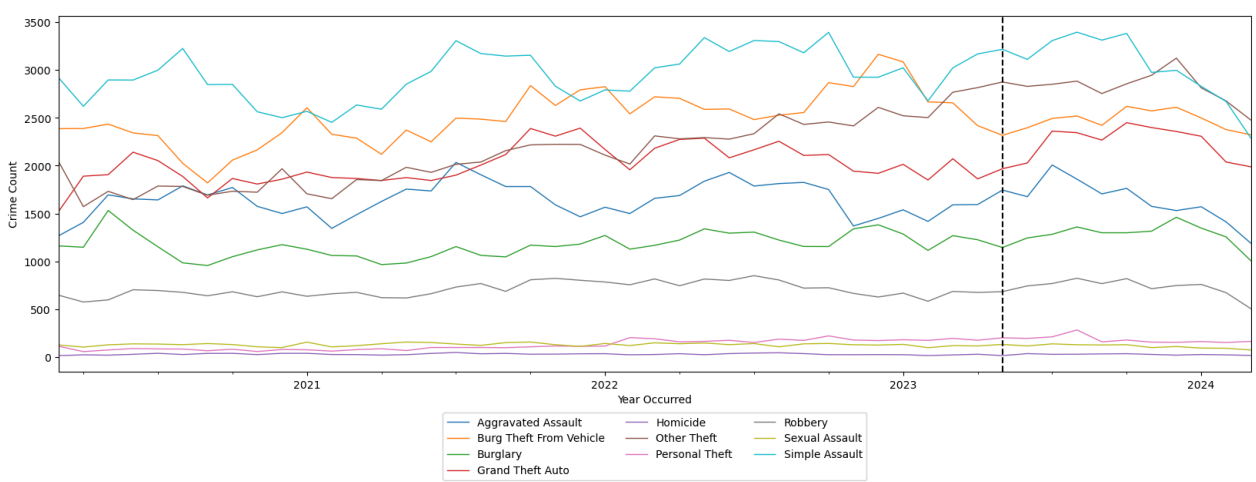


Figure 9: Time series plot for each major crime category over the COVID and Post-COVID time frame. The Y axis are crime count totals by month. The black line indicates the end of the COVID time frame

Figure 9 indicates a broad finding across the majority of categories. For most crimes, there is an initial suppressive effect that starts shortly after the March 2020 lockdown period, followed by a relatively quick escalation back to levels seen at the beginning of the period. The majority of crime types appear to be at their lowest around 2021, which could be a combined effect of seasonal fluctuation with COVID restrictions.

Equally interesting, is that most crime types observe a natural fluctuation that serves to obfuscate the effect of COVID changes. For example, over the entire frame, Simple Assault experiences four major dips over winter seasons. The 2021 COVID dip is the deepest, but overall the effect appears to have more to do with the time of year than the pandemic.

There are two clearer exceptions to the rule stated above, Burglary Theft from Vehicle (orange) and "Other Theft" (brown). Burglary Theft from Vehicle appears to be conducted at inverse seasons from the majority of other crime types, like Simple Assault. Additionally, vehicle theft counts vary, but over the period there is a positive trend, at least until the end COVID.

Type: Because COVID changed the routines of an entire society, we figured that the type of crime might have changed over the period as well. We expected that fewer homicides were committed during COVID as people were home more often. We have already seen through the time analysis that many types of crime did not appear to fluctuate solely because of COVID. It might make sense then that there was a relatively stable count of the types of crime committed as well

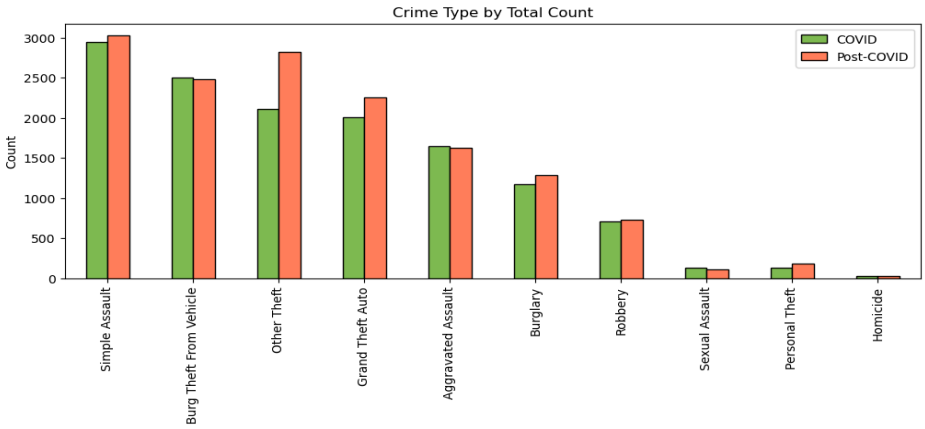


Figure 10: Count by crime type per month, normalized for both periods

For the most part, figure 10 confirms our new suspicion that most types of crime were conducted at a similar pace between both periods. In most cases, there were more crimes per month of a given category Post-COVID compared to during COVID. In the instances where COVID outpaced Post-COVID numbers, the margin was by a small number.

There are two clear outliers here as well. Other Theft, which was on the rise throughout both periods, increased by nearly 700 instances per month between the periods. Similarly, Grand Theft Auto increased in average cases per month in the Post-Covid timeframe. The Burglary category shows a noticeable increase as well, however it is unclear if this is due to regular variation or some other effect.

Time: As a final category under Question 2, we analyzed whether COVID played a role in the time of day that crime was conducted. Recall that during our initial exploration, we determined that Day of Week did not appear to be a significant factor in the type of crime committed. To explore further, we analyzed the hourly rate that crimes occurred during both the COVID and Post-COVID timeframes. Here, we highlight two examples, one where there does not seem to be a dramatic difference in time of crime, and another where the opposite is True.

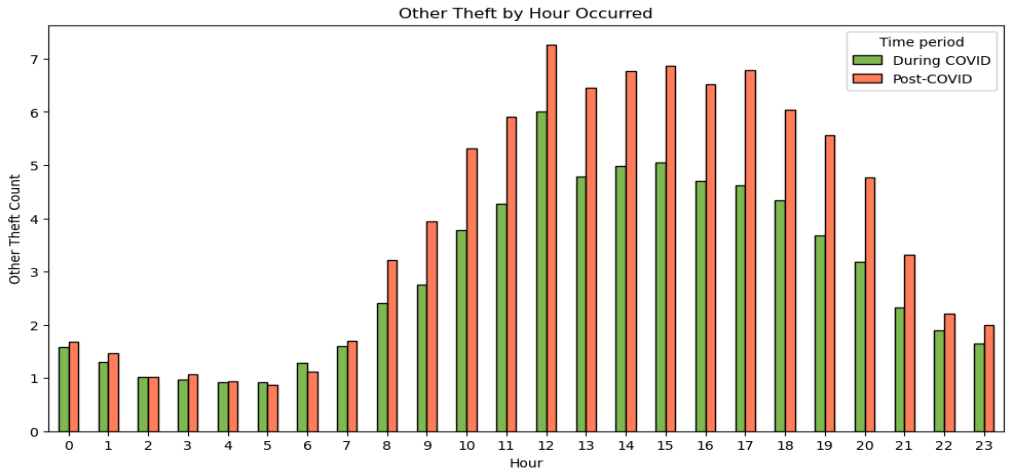


Figure 11: "Other Theft" count by the hour occurred. These values are normalized to be an average by day

In the case of Other Theft, COVID did not appear to change the time of day that the crime was committed, rather there seems to have been an overall suppression effect on the amount of crime that was committed, especially during "peak" times from 10 AM to 7 PM. Other plots have indicated that this type of crime was on the rise throughout the COVID period and that it was most severe in the Post-COVID period. It appears for this crime type there is no correlation between COVID and the time the crime was committed. Several other crime types fit a similar pattern, although none are as pronounced as for Other Theft.

However, the case of Homicides is different. Observe figure 12, a similar hourly plot of homicide count by the hour that the crime occurred.

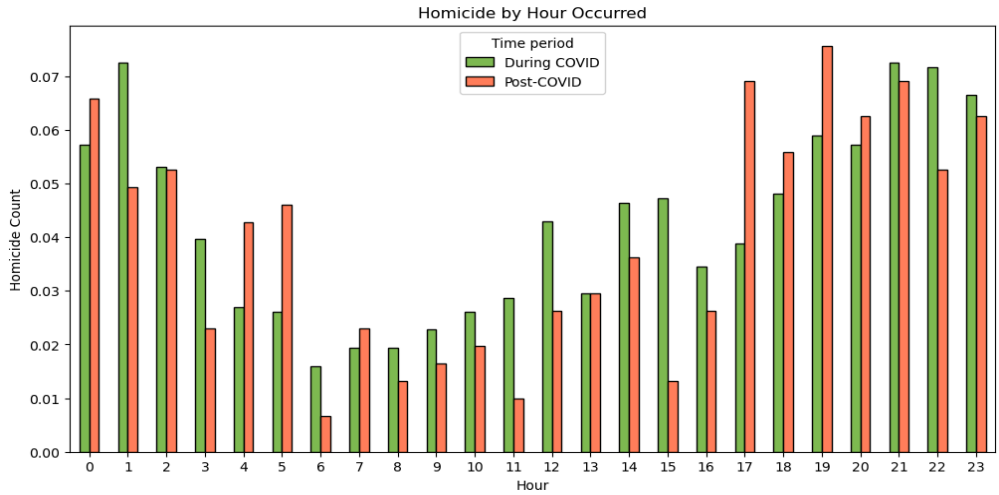


Figure 12: A plot of daily Homicide counts by the hour that they occurred. These values are normalized to be an average by day for both COVID and Post-COVID periods.

For Homicides in Los Angeles over the period, there is a shift in time of occurrence between the COVID and Post-COVID periods. During COVID, homicide was more likely to occur at almost all non-peak hours, 6 AM to 4 PM than in Post-COVID. Additionally, during many evening Peak hours, 21 PM to 03 AM, COVID rates matched or exceeded Post-COVID rates.

We are not able to determine that COVID resulted in the effect of the shift in homicide occurrence times, but it is clear that there is a difference between both periods. We did not observe a clear timing shift for any other crime category in our analysis.

In summary of Question 2, we observe that the frequency of most crimes did not appear to be correlated to COVID, with exceptions being Burglary from Vehicle and Other theft. Similarly, Other Theft and Grand theft auto increased in frequency in the Post-COVID period, while most other crimes

remained stable. Finally, we observed that the timing of crime changed for Homicide, but was the same for all other crimes. Notably, Other Theft occurred at the same time for both periods, but increased in frequency during peak times in the Post-COVID period.

Question 3: (SRIYA)

How did crime against certain victim demographics change during COVID versus post- COVID?

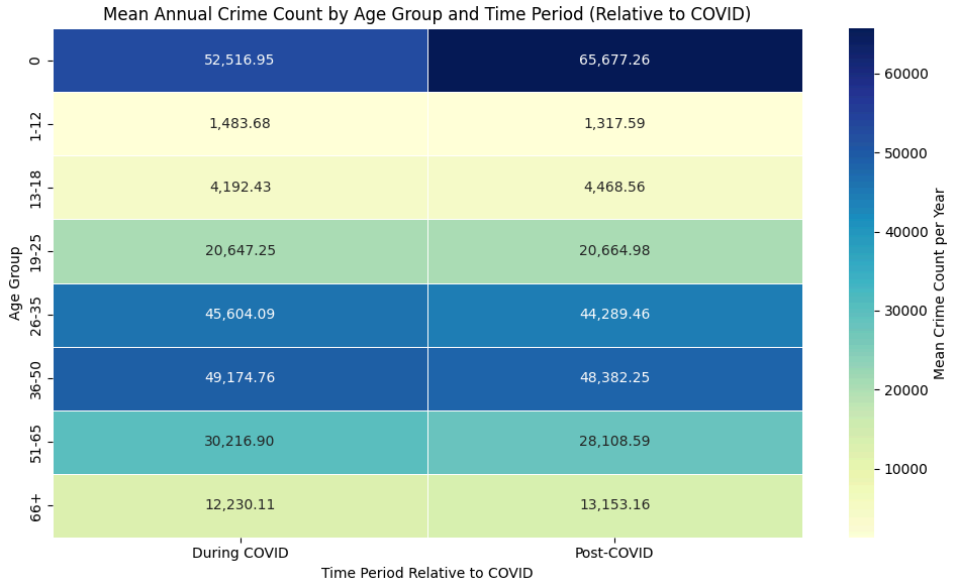


Figure 13: Heat Map for Mean Annual Crime Count by Age Group and Time Period (comparing the COVID and Post-COVID time frames). The left column represents the crime counts during COVID, and the right column represents the crime counts for Post - COVID. The Y-axis showcases the Age Groups and the normalized crime counts to analyze trends between age and time period.

We found a significant increase by 25% of victimless crimes (age 0) from during COVID to post COVID! We also saw a 6.6% increase to crimes where the victim was between 13 to 18 and 7.5% increases to crimes where the victim was 66+. This displays a trend of increased crime against teens and the elderly. However, there was also an 11.2% decrease of victims between 1 to 12.

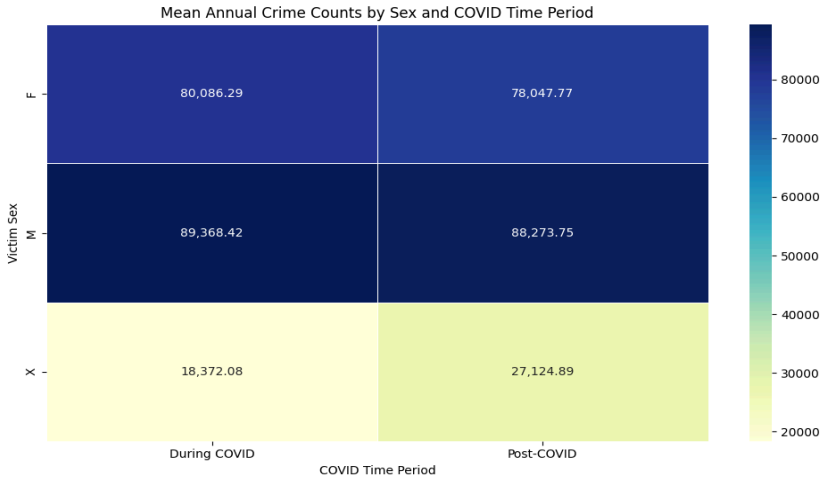


Figure 14: Heat Map for Mean Annual Crime Count by Sex Group and Time Period (comparing the COVID and Post-COVID time frames). The left column represents the crime counts during COVID, and the right column represents the crime counts for Post - COVID. The Y-axis showcases the Sex Groups and the normalized crime counts to analyze trends between age and time period.

Most notably, we saw an increase by 46% of crimes with unknown sex (category X) during

COVID to post-COVID. Once again, this coincides with our trend of increased victimless crimes Post-COVID. Furthermore, we noticed a miniscule decrease between 1 to 2.5% in crime rates against males and females, going against our hypothesis that COVID had a large impact on crime rate.

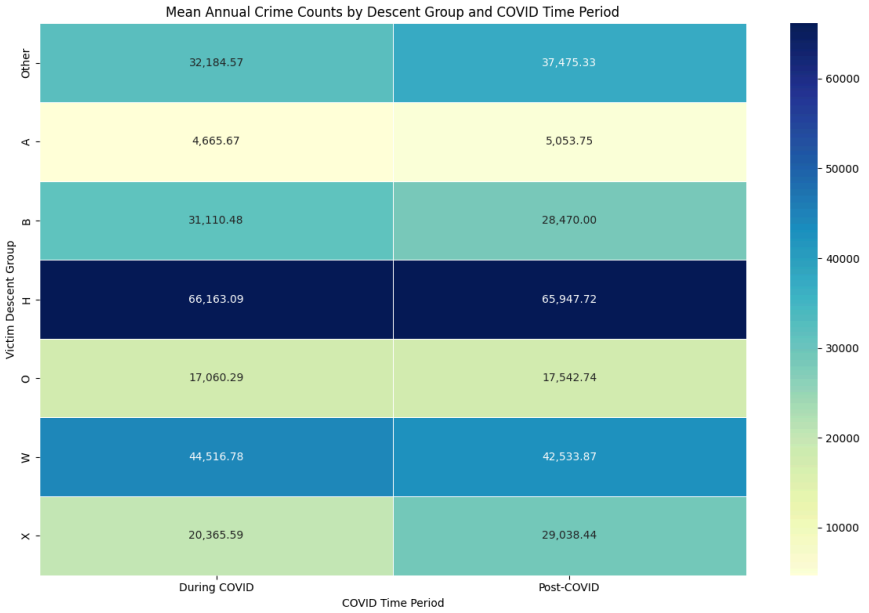
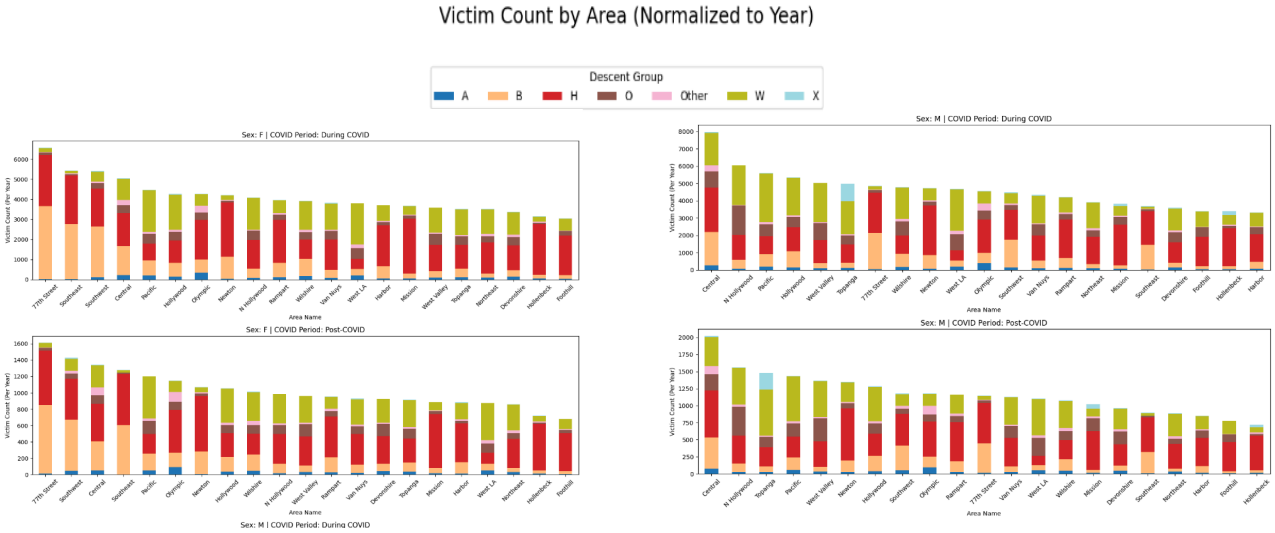


Figure 15: Heat Map for Mean Annual Crime Count by Descent Group and Time Period (comparing the COVID and Post-COVID time frames). The left column represents the crime counts during COVID, and the right column represents the crime counts for Post - COVID. The Y-axis showcases the Descent Groups and the normalized crime counts to analyze trends between age and time period.

Our analysis shows an Increase of crime by 8.3% against other Asians (Not Chinese, Japanese, Cambodian, Filipino, or Asian Indian). This is an interesting observation as it coincides with national news during and post covid of increased Asian American hate crimes when 49% of AAPI adults reported experiencing hate due to race, ethnicity or nationality[1].

Additionally, we see a 42.6% increase in crimes where the descent is unknown, which is answered by victimless crimes.



[1] 2023, *State of Safety for Asian American and Pacific Islander Communities, stopaapihate.org*

Figure 16: Bar Graph for Victim Crime Count by Area (Normalized to Year Using Mean) by Descent Group and Time Period (comparing the COVID and Post-COVID time frames). The left visual displays two charts for Sex: Female with COVID on top and Post- COVID on the bottom. The right visual displays two charts for Sex: Male with COVID on top and Post-COVID on the bottom. The x-axes show the Area and the y-axes showcase the Victim Count normalized to year.

This bar chart shows the distribution of victim descent count by area, separate into two different graphs for male and female. We see that Newton and mission have a higher relative Hispanic victim descent for both males and females. On the other hand, Topanga has a higher rate of Hispanic victim descent for males, but a higher rate of White victim descent for females.

Most importantly, for females there seems to be a higher rate of Hispanic victims in the top five areas of crime counts, but for men there seems to be a higher rate of White victims in the top five areas of crime counts.

Conclusion

- **Hypothesis 1:** Only a half of the area's crime percentage increased in post–COVID.

  Not all the areas crime major crime type changed, most of them remain the same except 5 areas

- **Hypothesis 2:** How did crime change between the two periods? Did crime change in frequency, type, or time?
  - Some types of Crime, like Theft and Burglary Theft from vehicle, did increase in frequency.
  - Other Theft and Grand theft Auto substantially increased in the Post-COVID period
  - Homicide changed time of occurrence between the two periods. Other crimes remained stable.
- **Hypothesis 3:** How did crime against certain victim demographics change between the COVID and post-COVID periods?
  - Post - COVID, we see a trend of higher victimless crimes (e.g. traffic violations and theft).
  - Post - COVID, We see an increase of crime against those in the other Asian category.
  - There was a miniscule decrease in crime against men and women (between 1 to 2.5%).
  - For females, there is a higher rate of hispanic victims in the top five crime count areas. For males, there is a higher rate of white victims in the top five crime count areas.